Diffusion

Exam code: 0620 & 0971

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Diffusion

- Diffusion occurs in gases and liquids, due to the random motion of their particles
- It is where particles move from an area of **high concentration** to an area of **low concentration**
- Eventually the concentration of particles is **even** as the particles are evenly spread throughout the available space
- Diffusion happens on its own and no energy input is required
  - Although, it occurs faster at higher temperatures because the particles have more kinetic energy

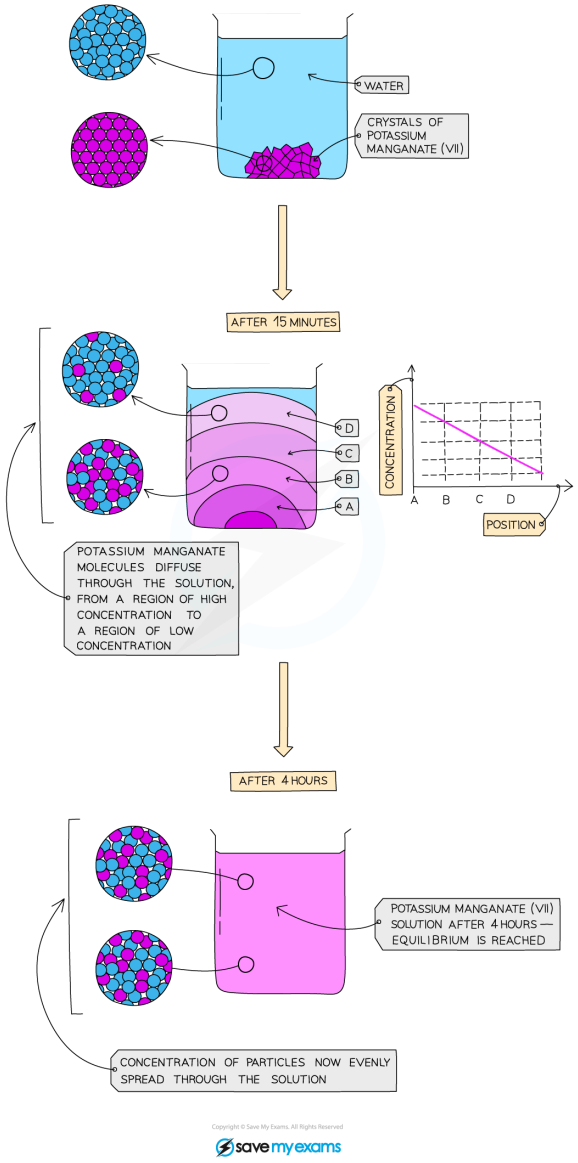
Diffusion in liquids

- Potassium manganate(VII), $KMnO_4$, in water is a typical demonstration of diffusion in liquids:

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

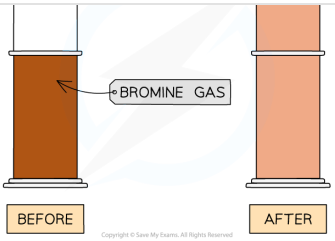


Diffusion of potassium manganate(VII), $KMnO_4$, in water. After a few hours, the concentration of $KMnO_4$ is the same throughout the solution

Diffusion in gases

- Diffusion is **faster** in gases than in liquids
  - This is because gaseous particles have more energy and move quicker than liquid particles
- For example, the diffusion of bromine gas and air:

Help me understand



Bromine diffuses until it is evenly spread throughout the container

- At the start, the **orange-brown** bromine gas is an area of high concentration
- It diffuses from a high to low concentration
- After 5 minutes, the bromine gas will have diffused from the bottom jar until it is **evenly spread** throughout both jars
  - The same can be said for the air, although it is less obvious as it is colourless

Examiner Tips and Tricks

When you are talking about diffusion, you should (where appropriate) include:

- Ideas about areas of high and low concentration
- The energy / movement of the particles

Diffusion & molecular mass

Extended tier only

How molecular mass affects diffusion

- At the same temperature, different gases do not diffuse at the same rate.
- This is due to the difference in their **relative molecular masses**
- Gases with a lower relative molecular mass are "lighter" which means that they:
  - Travel faster
  - Travel further in the same amount of time
- The reverse argument is true for gases with a high relative molecular mass, they:
  - Travel slower
  - Do not travel as far in the same amount of time
- For example, the reaction between ammonia and hydrogen chloride

- Ammonia gas and hydrogen chloride gas react together to form solid ammonium chloride

$$NH_3\,(g) + HCl\,(g) \rightarrow NH_4Cl\,(s)$$

- The M_r of ammonia is 17, while the M_r of hydrogen chloride is 36.5
  - This means that ammonia will travel faster and further than hydrogen chloride
  - Therefore, the ammonium chloride will form nearer to the hydrogen chloride
    - This is seen as a white "smoke" ring inside the gas tube

Ammonia molecules have less mass than HCl molecules so they diffuse faster and the product forms closer to the HCl end

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